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We here at CertyIQ eventually got enough of the industry's greedy exam paid for. Our team of IT professionals comes with years of experience in the IT industry Prior to training CertyIQ we worked in test areas where we observed the horrors of the paywall exam preparation system.

The misuse of the preparation system has left our team disillusioned. And for that reason, we decided it was time to make a difference. We had to make In this way, CertyIQ was created to provide quality materials without stealing from everyday people who are trying to make a living.

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We have developed a very scalable solution using which we are able to solve 400+ doubts every single day with an average rating of 4.8 out of 5.

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John

October 19, 2022



Thanks you so much for your help. I scored 972 in my exam today. More than 90% were from your PDFs!

Dana

September 04, 2022



Thanks a lot for this updated AZ-900 Q&A. I just passed my exam and got 974, I followed both of your Az-900 videos and the 6 PDF, the PDFs are very much valid, all answers are correct. Could you please create a similar video/PDF for DP900, your content/PDF's is really awesome. The team did a really good job. Thank You 😊.

Ahamed Shibly

2 months ago



Customer support is really fast and helpful, I just finished my exam and this video along with the 6 PDF helped me pass! Definitely recommend getting the PDFs. Thank you!

October 22, 2022



Passed my exam today with 891 marks. Out of 52 questions, 51 were from certyiq PDFs including Contoso case study. Thank You certyiq team!

Henry Rome

2 months ago



These questions are real and 100 % valid. Thank you so much for your efforts, also your 4 PDFs are awesome, I passed the DP900 exam on 1 Sept. With 968 marks. Thanks a lot, buddy!

Esmaria

2 months ago



Simple easy to understand explanations. To anyone out there wanting to write AZ900, I highly recommend 6 PDF's. Thank you so much, appreciate all your hard work in having such great content. Passed my exam Today - 3 September with 942 score.

Google

(Associate Data Practitioner)

Google Cloud Certified - Associate Data Practitioner

Total: **72 Questions**

Link: <https://certyiq.com/papers/google/associate-data-practitioner>

Question: 1

Your retail company wants to predict customer churn using historical purchase data stored in BigQuery. The dataset includes customer demographics, purchase history, and a label indicating whether the customer churned or not. You want to build a machine learning model to identify customers at risk of churning. You need to create and train a logistic regression model for predicting customer churn, using the customer_data table with the **churned column as the target label**. Which BigQuery ML query should you use?

A.

```
CREATE OR REPLACE MODEL churn_prediction_model
OPTIONS(model_type='logistic_reg') AS
SELECT *
FROM customer_data;
```

B.

```
CREATE OR REPLACE MODEL churn_prediction_model
OPTIONS(model_type='logistic_reg') AS
SELECT * EXCEPT(churned),
       churned AS label
FROM customer_data;
```

C.

```
CREATE OR REPLACE MODEL churn_prediction_model
OPTIONS(model_type='logistic_reg') AS
SELECT * EXCEPT(churned)
FROM customer_data;
```

D.

```
CREATE OR REPLACE MODEL churn_prediction_model
OPTIONS(model_type='logistic_reg') AS
SELECT churned as label
FROM customer_data;
```

Answer: B**Explanation:**

B.Option B is best because it correctly selects all columns as features except the churned column, and then explicitly designates the churned column as the label for BigQuery ML, which is required for training. Option A (SELECT *) is incorrect because while it includes the churned column, it doesn't explicitly define it as the label. Option C (SELECT * EXCEPT(churned)) is incorrect because it excludes the target churned column, which is essential for training a supervised model. Option D (SELECT churned as label) is incorrect because it only selects the label and excludes all feature columns needed for prediction. Therefore, Option B is the only query that correctly sets up the data for BigQuery ML logistic regression training.

Question: 2

CertyIQ

Your company has several retail locations. Your company tracks the **total number of sales made at each location each day**. You want to use SQL to calculate the **weekly moving average of sales by location** to identify trends for each store. Which query should you use?

A.
`SELECT store_id, date, total_sales, AVG(total_sales) OVER (
PARTITION BY store_id
ORDER BY total_sales RANGE BETWEEN 6 PRECEDING AND CURRENT ROW) as rolling_avg
FROM store_sales_daily`

B.
`SELECT store_id, date, total_sales, AVG(total_sales)
OVER (
PARTITION BY date ORDER BY store_id ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) as rolling_avg
FROM store_sales_daily`

C.
`SELECT store_id, date, total_sales, AVG(total_sales)
OVER (
PARTITION BY store_id
ORDER BY date ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) as rolling_avg
FROM store_sales_daily`

D.
`SELECT store_id, date, total_sales, AVG(total_sales)
OVER (
PARTITION BY total_sales
ORDER BY date RANGE BETWEEN 6 PRECEDING AND CURRENT ROW) as rolling_avg
FROM store_sales_daily`

Answer: C

Explanation:

C. Option C is best because it correctly partitions by store_id to calculate the moving average for each store individually, and orders by date to ensure the moving average is calculated chronologically over the preceding 7 days (including the current day, thus 6 preceding). ROWS BETWEEN 6 PRECEDING AND CURRENT ROW then correctly defines the 7-day window for the average. Option A is incorrect because it orders by total_sales

instead of date, making the rolling average based on sales value order, not time, which is illogical for a weekly trend. Option B is incorrect because it partitions by date instead of store_id, calculating a daily moving average across all stores, not per store. Option D is incorrect because it partitions by total_sales, which is nonsensical for analyzing trends by location, and orders by date within that illogical partition. Therefore, Option C is the only query that correctly calculates a weekly moving average of sales by store location.

Question: 3

CertyIQ

Your company is building a near **real-time streaming pipeline** to process JSON telemetry data from small appliances. You need to process messages arriving at a Pub/Sub topic, **capitalize letters in the serial number field**, and write results to BigQuery. You want to use **a managed service** and write a **minimal amount of code for underlying transformations**. What should you do?

- A. Use a Pub/Sub to BigQuery subscription, write results directly to BigQuery, and schedule a transformation query to run every five minutes.
- B. Use a Pub/Sub to Cloud Storage subscription, write a Cloud Run service that is triggered when objects arrive in the bucket, performs the transformations, and writes the results to BigQuery.
- C. Use the “Pub/Sub to BigQuery” Dataflow template with a UDF, and write the results to BigQuery.
- D. Use a Pub/Sub push subscription, write a Cloud Run service that accepts the messages, performs the transformations, and writes the results to BigQuery.

Answer: C

Explanation:

C. Use the “Pub/Sub to BigQuery” Dataflow template with a UDF. Option C is best because Dataflow templates are managed, serverless, and designed for streaming Pub/Sub data to BigQuery. UDFs allow minimal code for transformations within the pipeline. Option A (Pub/Sub to BigQuery + scheduled query) is incorrect because scheduled queries are not real-time transformations. Option B (Pub/Sub to Cloud Storage + Cloud Run) is incorrect because it adds unnecessary complexity with Cloud Storage as an intermediary and is not truly streaming. Option D (Pub/Sub push + Cloud Run) is incorrect because while real-time, it requires more code in Cloud Run than using a Dataflow UDF and is less purpose-built for data pipelines than Dataflow. Therefore, Option C, Dataflow template with UDF, is the best balance of managed service, minimal code, and near real-time streaming.

Question: 4

CertyIQ

You want to process and load a daily sales CSV file stored in Cloud Storage into BigQuery for downstream reporting. You need to quickly build a **scalable data pipeline that transforms the data while providing insights into data quality issues**. What should you do?

- A. Create a batch pipeline in Cloud Data Fusion by using a Cloud Storage source and a BigQuery sink.
- B. Load the CSV file as a table in BigQuery, and use scheduled queries to run SQL transformation scripts.
- C. Load the CSV file as a table in BigQuery. Create a batch pipeline in Cloud Data Fusion by using a BigQuery source and sink.
- D. Create a batch pipeline in Dataflow by using the Cloud Storage CSV file to BigQuery batch template.

Answer: A

Explanation:

- A. Create a batch pipeline in Cloud Data Fusion by using a Cloud Storage source and a BigQuery sink.

Cloud Data Fusion is a fully managed, scalable data integration service that makes building and managing pipelines straightforward.

By using a Cloud Storage source, you can ingest the daily CSV file easily.

With a BigQuery sink, you can load and store the transformed data directly in BigQuery for downstream reporting.

Cloud Data Fusion also provides tools to monitor and detect data quality issues, aligning with your need for insights into data quality.

Question: 5

CertyIQ

You manage a Cloud Storage bucket that stores temporary files created during data processing. These temporary files are only needed for seven days, **after which they are no longer needed**. **To reduce storage costs and keep your bucket organized**, you want **to automatically delete these files once they are older than seven days**. What should you do?

- A. Set up a Cloud Scheduler job that invokes a weekly Cloud Run function to delete files older than seven days.
- B. Configure a Cloud Storage lifecycle rule that automatically deletes objects older than seven days.
- C. Develop a batch process using Dataflow that runs weekly and deletes files based on their age.
- D. Create a Cloud Run function that runs daily and deletes files older than seven days.

Answer: B

Explanation:

B. Configure a Cloud Storage lifecycle rule that automatically deletes objects older than seven days.

Since the goal is to automate the deletion of temporary files older than seven days, the best approach is to use a built-in Cloud Storage feature that requires minimal effort and is highly efficient.

Cloud Storage lifecycle rules allow you to define policies to automatically delete, archive, or change the storage class of objects based on age.

You can set a rule to delete objects after seven days, eliminating the need for manual intervention or additional services.

It is cost-effective, serverless, and requires no maintenance.

Question: 6

CertyIQ

You work for a healthcare company that has a large on-premises data system containing patient records with personally identifiable information (PII) such as names, addresses, and medical diagnoses. You need a **standardized managed solution** that de-identifies PII across all your data feeds prior to **ingestion to Google Cloud**. What should you do?

- A. Use Cloud Run functions to create a serverless data cleaning pipeline. Store the cleaned data in BigQuery.
- B. Use Cloud Data Fusion to transform the data. Store the cleaned data in BigQuery.
- C. Load the data into BigQuery, and inspect the data by using SQL queries. Use Dataflow to transform the data and remove any errors.
- D. Use Apache Beam to read the data and perform the necessary cleaning and transformation operations. Store the cleaned data in BigQuery.

Answer: B

Explanation:

B. Use Cloud Data Fusion to transform the data. Store the cleaned data in BigQuery.

The goal is to de-identify personally identifiable information (PII) across all data feeds before ingestion into Google Cloud, using a standardized managed solution.

Cloud Data Fusion is a fully managed ETL (Extract, Transform, Load) service that allows for data transformation at scale with built-in connectors for both on-premises and cloud data sources.

Cloud Data Fusion integrates with Cloud DLP (Data Loss Prevention), which provides automatic PII detection and de-identification before storing data in BigQuery.

It offers pre-built data processing pipelines that allow for standardized PII redaction, masking, or tokenization before ingestion into Google Cloud.

No manual intervention is required, making it scalable and efficient.

Question: 7

CertyIQ

You manage a large amount of data in Cloud Storage, including raw data, processed data, and backups. Your organization is subject to **strict compliance regulations** that mandate **data immutability** for specific data types. You want to use an **efficient process to reduce storage costs while ensuring that your storage strategy meets retention requirements**. What should you do?

- A. Configure lifecycle management rules to transition objects to appropriate storage classes based on access patterns. Set up Object Versioning for all objects to meet immutability requirements.
- B. Move objects to different storage classes based on their age and access patterns. Use Cloud Key Management Service (Cloud KMS) to encrypt specific objects with customer-managed encryption keys (CMEK) to meet immutability requirements.
- C. Create a Cloud Run function to periodically check object metadata, and move objects to the appropriate storage class based on age and access patterns. Use object holds to enforce immutability for specific objects.
- D. Use object holds to enforce immutability for specific objects, and configure lifecycle management rules to transition objects to appropriate storage classes based on age and access patterns.

Answer: D

Explanation:

D. Use object holds to enforce immutability for specific objects, and configure lifecycle management rules to transition objects to appropriate storage classes based on age and access patterns.

Object holds are a feature in Cloud Storage specifically designed to enforce immutability. With a hold in place, an object cannot be deleted or overwritten until the hold is released, ensuring compliance with strict data retention requirements.

Lifecycle management rules help automatically transition objects to cost-effective storage classes (e.g., Nearline, Coldline, or Archive) based on their age or access patterns, reducing storage costs efficiently.

Question: 8

CertyIQ

You work for an ecommerce company that has a **BigQuery dataset** that contains customer purchase history, demographics, and website interactions. You need to build a machine learning (**ML**) model to predict which

customers are most likely to make a purchase in the next month. You have **limited engineering resources and need to minimize the ML expertise required for the solution**. What should you do?

- A. Use BigQuery ML to create a logistic regression model for purchase prediction.
- B. Use Vertex AI Workbench to develop a custom model for purchase prediction.
- C. Use Colab Enterprise to develop a custom model for purchase prediction.
- D. Export the data to Cloud Storage, and use AutoML Tables to build a classification model for purchase prediction.

Answer: A

Explanation:

A. Use BigQuery ML to create a logistic regression model for purchase prediction.

BigQuery ML allows you to build and train machine learning models directly within BigQuery using simple SQL commands, minimizing the need for advanced ML expertise.

It integrates seamlessly with your existing dataset in BigQuery, eliminating the need to move data around, which saves time and reduces complexity.

Logistic regression is an effective and efficient algorithm for binary classification tasks, such as predicting whether or not a customer will make a purchase.

This approach is quick, scalable, and tailored for teams with limited engineering resources.

Question: 9

CertyIQ

You are designing a **pipeline to process data files that arrive in Cloud Storage** by 3:00 am each day. Data processing is performed in stages, where the **output of one stage becomes the input of the next**. Each stage takes a long time to run. Occasionally **a stage fails, and you have to address the problem**. You need to ensure that the **final output is generated as quickly as possible**. What should you do?

- A. Design a Spark program that runs under Dataproc. Code the program to wait for user input when an error is detected. Rerun the last action after correcting any stage output data errors.
- B. Design the pipeline as a set of PTransforms in Dataflow. Restart the pipeline after correcting any stage output data errors.
- C. Design the workflow as a Cloud Workflow instance. Code the workflow to jump to a given stage based on an input parameter. Rerun the workflow after correcting any stage output data errors.
- D. Design the processing as a directed acyclic graph (DAG) in Cloud Composer. Clear the state of the failed task after correcting any stage output data errors.

Answer: D

Explanation:

D. Design the processing as a directed acyclic graph (DAG) in Cloud Composer. Clear the state of the failed task after correcting any stage output data errors. Cloud Composer (Apache Airflow) is specifically designed for orchestrating complex data pipelines, provides robust error handling with task-level rerun capabilities for efficient recovery, and offers a fully managed and scheduled environment, making it the most suitable choice for ensuring the final output is generated as quickly as possible in the face of occasional stage failures. Options A, B, and C are less efficient due to manual intervention, less granular error recovery, or lack of dedicated workflow orchestration features.

Question: 10

Another team in your organization is requesting access to a BigQuery dataset. You need to **share the dataset with the team while minimizing the risk of unauthorized copying of data**. You also want to create **a reusable framework** in case you need **to share this data with other teams in the future**. What should you do?

- A. Create authorized views in the team's Google Cloud project that is only accessible by the team.
- B. Create a private exchange using Analytics Hub with data egress restriction, and grant access to the team members.
- C. Enable domain restricted sharing on the project. Grant the team members the BigQuery Data Viewer IAM role on the dataset.
- D. Export the dataset to a Cloud Storage bucket in the team's Google Cloud project that is only accessible by the team.

Answer: A

Explanation:

A. Create authorized views in the team's Google Cloud project that is only accessible by the team.

The goal is to share a BigQuery dataset securely while:

Minimizing the risk of unauthorized data copying.

Creating a reusable framework for future data-sharing needs.

Authorized views in BigQuery allow controlled access to specific columns or rows instead of sharing the full dataset.

The underlying dataset remains protected, and users can only query through the authorized view, preventing direct exports or full access.

This is a scalable and reusable framework for future team data-sharing needs.

Question: 11

Your company has developed a website that allows users to upload and share video files. These files are most frequently accessed and shared when they are initially uploaded. Over time, the **files are accessed and shared less frequently**, although some old video files may remain very popular. You need to design a storage system that is **simple and cost-effective**. What should you do?

- A. Create a single-region bucket with Autoclass enabled.
- B. Create a single-region bucket. Configure a Cloud Scheduler job that runs every 24 hours and changes the storage class based on upload date.
- C. Create a single-region bucket with custom Object Lifecycle Management policies based on upload date.
- D. Create a single-region bucket with Archive as the default storage class.

Answer: A

Explanation:

A. Create a single-region bucket with Autoclass enabled.

Your goal is to design a simple and cost-effective storage system that adapts to changing access patterns:

Frequent access at the start (hot storage needed).

Less frequent access over time (cold storage preferred).

Some old files may remain popular (must stay in higher-performance storage if needed).

Autoclass automatically moves objects between storage classes (Standard, Nearline, and Coldline) based on access patterns without requiring manual policies or scripting.

It eliminates complexity, as you don't need lifecycle rules or scheduled jobs.

It ensures optimal cost savings while keeping popular files readily accessible.

No retrieval fees when accessing objects in colder storage classes.

Question: 12

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You recently inherited a task for **managing Dataflow streaming pipelines** in your organization and noticed that proper access had not been provisioned to you. You need to request a Google-provided IAM role so you can **restart the pipelines**. You need to follow the principle of least privilege. What should you do?

- A. Request the Dataflow Developer role.
- B. Request the Dataflow Viewer role.
- C. Request the Dataflow Worker role.
- D. Request the Dataflow Admin role.

Answer: A

Explanation:

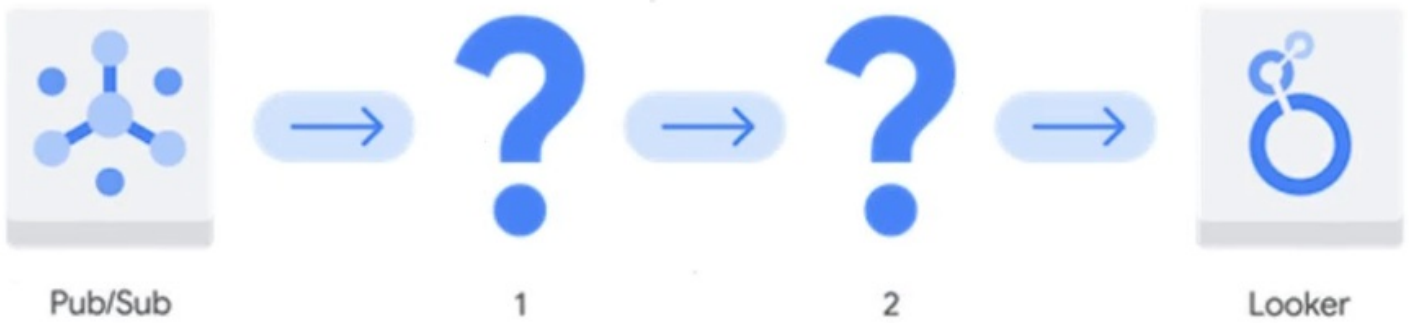
A. Request the Dataflow Developer role. Requesting the Dataflow Developer role is optimal because it grants the necessary permissions to restart Dataflow pipelines, aligning with the user's need, while adhering to the principle of least privilege by providing a focused set of permissions. Option B (Dataflow Viewer) is insufficient as it only provides read-only access and lacks the permission to restart pipelines. Option C (Dataflow Worker) is incorrect as it is intended for service accounts, not user access, and doesn't grant pipeline management permissions. Option D (Dataflow Admin) is excessive, granting far more permissions than needed for simply restarting pipelines, thus violating the principle of least privilege. Therefore, Option A is the clear and correct choice for requesting the minimum necessary IAM role to restart Dataflow pipelines.

Question: 13

CertyIQ

You need to create a new data pipeline. You want a serverless solution that meets the following requirements:

- Data is streamed from Pub/Sub and is processed in real-time.
- Data is transformed before being stored.
- Data is stored in a location that will allow it to be analyzed with SQL using Looker.



Which Google Cloud services should you recommend for the pipeline?

- A.1. Dataproc Serverless
- 2. Bigtable
- B.1. Cloud Composer
- 2. Cloud SQL for MySQL
- C.1. BigQuery
- 2. Analytics Hub
- D.1. Dataflow
- 2. BigQuery

Answer: D

Explanation:

D. 1. Dataflow 2. BigQuery.

You need to recommend Google Cloud services for a data pipeline, which typically involves:

Streaming or batch data processing.

Efficient storage and querying for analytics.

Dataflow is a serverless, fully managed service designed for ETL (Extract, Transform, Load) and data processing. It supports:

Real-time (streaming) and batch data processing.

Integration with BigQuery, Pub/Sub, and Cloud Storage.

Apache Beam-based pipelines for scalability and efficiency.

BigQuery is a fully managed, serverless data warehouse for:

Fast analytics on large datasets.

Seamless integration with Dataflow, Cloud Storage, and other services.

SQL-based querying without infrastructure management.

Dataflow + BigQuery is the ideal solution for an efficient and scalable data pipeline in Google Cloud.

Question: 14

CertyIQ

Your team wants to create a monthly report to analyze inventory data that is **updated daily**. You need to **aggregate the inventory counts by using only the most recent month of data**, and save the results to be used in a Looker Studio dashboard. What should you do?

- A. Create a materialized view in BigQuery that uses the SUM() function and the DATE_SUB() function.
- B. Create a saved query in the BigQuery console that uses the SUM() function and the DATE_SUB() function. Re-run the saved query every month, and save the results to a BigQuery table.
- C. Create a BigQuery table that uses the SUM() function and the _PARTITIONDATE filter.
- D. Create a BigQuery table that uses the SUM() function and the DATE_DIFF() function.

Answer: A

Explanation:

A. Create a materialized view in BigQuery that uses the SUM() function and the DATE_SUB() function. Materialized views provide the efficiency of pre-computation, automatic updates, and are directly usable in Looker Studio, making them the ideal choice for creating an automated and performant monthly inventory report. Options B, C, and D are either manual, technically inaccurate in their description, or less efficient than leveraging the power of materialized views for this reporting task.

Question: 15

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You have a BigQuery dataset containing sales data. This data is **actively queried for the first 6 months**. After that, the data is not queried but **needs to be retained for 3 years for compliance reasons**. You need to implement a data **management strategy that meets access and compliance requirements, while keeping cost and administrative overhead to a minimum**. What should you do?

- A. Use BigQuery long-term storage for the entire dataset. Set up a Cloud Run function to delete the data from BigQuery after 3 years.
- B. Partition a BigQuery table by month. After 6 months, export the data to Coldline storage. Implement a lifecycle policy to delete the data from Cloud Storage after 3 years.
- C. Set up a scheduled query to export the data to Cloud Storage after 6 months. Write a stored procedure to delete the data from BigQuery after 3 years.
- D. Store all data in a single BigQuery table without partitioning or lifecycle policies.

Answer: B

Explanation:

B. Partition a BigQuery table by month. After 6 months, export the data to Coldline storage. Implement a lifecycle policy to delete the data from Cloud Storage after 3 years. Option B provides the optimal balance of meeting the 3-year data retention requirement, achieving excellent cost optimization by tiering data to Coldline, and maintaining reasonable administrative overhead. Option C is more complex and less clear on overall retention and cost due to data duplication. Option D is the simplest administratively but provides the worst cost optimization. Option A is not cost-optimized and has moderate overhead. Therefore, Option B is the most practical and well-rounded solution for this data management scenario.

Question: 16

CertyIQ

You have created a LookML model and dashboard that **shows daily sales metrics for five regional managers to use**. You want to ensure that the **regional managers can only see sales metrics specific to their region**. You need an **easy-to-implement** solution. What should you do?

- A. Create a sales_region user attribute, and assign each manager's region as the value of their user attribute. Add an access_filter Explore filter on the region_name dimension by using the sales_region user attribute.
- B. Create five different Explores with the sql_always_filter Explore filter applied on the region_name dimension. Set each region_name value to the corresponding region for each manager.

C.Create separate Looker dashboards for each regional manager. Set the default dashboard filter to the corresponding region for each manager.

D.Create separate Looker instances for each regional manager. Copy the LookML model and dashboard to each instance. Provision viewer access to the corresponding manager.

Answer: A

Explanation:

A. Create a sales_region user attribute, and assign each manager's region as the value of their user attribute. Add an access_filter Explore filter on the region_name dimension by using the sales_region user attribute. This option is optimal because it directly leverages Looker's built-in features for row-level security (Access Filters) and user personalization (User Attributes), making it easy to implement, scalable, maintainable, and secure. Option B (Separate Explores) is inefficient and less maintainable. Option C (Separate Dashboards) does not provide data security and is easily bypassed. Option D (Separate Looker Instances) is extreme overkill, costly, and impractical. Therefore, Option A is the clear and correct choice for an easy-to-implement and secure solution.

Question: 17

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You need to design a data pipeline that **ingests data** from CSV, Avro, and Parquet files into Cloud Storage. The data includes raw user input. You need to **remove all malicious SQL** injections before **storing the data** in BigQuery. Which data manipulation methodology should you choose?

- A.EL
- B.ELT
- C.ETL
- D.ETLT

Answer: C

Explanation:

C. ETL (Extract, Transform, Load)

You need to remove malicious SQL injections before storing the data in BigQuery. This requires data transformation before loading it into BigQuery, which aligns with the ETL methodology:

Extract .

Ingest data from CSV, Avro, and Parquet files stored in Cloud Storage.

Transform .

Remove malicious SQL injections before storing data.

Perform data cleansing, validation, and sanitization.

Load .

Load cleaned and structured data into BigQuery.

Question: 18

CertyIQ

You are working with a large dataset of customer reviews stored in Cloud Storage. The dataset contains several **inconsistencies, such as missing values, incorrect data types, and duplicate entries**. You need to **clean the data** to ensure that it is accurate and consistent before using it for analysis. What should you do?

- A. Use the **PythonOperator** in Cloud Composer to clean the data and load it into BigQuery. Use SQL for analysis.
- B. Use BigQuery to batch load the data into BigQuery. Use SQL for cleaning and analysis.
- C. Use Storage Transfer Service to move the data to a different Cloud Storage bucket. Use event triggers to invoke Cloud Run functions to load the data into BigQuery. Use SQL for analysis.
- D. Use Cloud Run functions to clean the data and load it into BigQuery. Use SQL for analysis.

Answer: A

Explanation:

A. Use the Python Operator in Cloud Composer to clean the data and load it into Big Query. Use SQL for analysis.

Cloud Composer: As a managed orchestration service built on Apache Airflow, Cloud Composer is excellent for managing and automating complex workflows. If your dataset requires multiple sequential cleaning steps that involve different tools or services, Cloud Composer can orchestrate these tasks efficiently.

PythonOperator: This allows you to write custom Python scripts for advanced data cleaning operations that may not be easily achievable with SQL alone. For instance, you might use Python libraries like pandas to handle missing values, reformat data, or detect duplicates.

Question: 19

CertyIQ

Your retail organization stores sensitive application usage data in Cloud Storage. You need to **encrypt the data without the operational overhead of managing encryption keys**. What should you do?

- A. Use Google-managed encryption keys (GMEK).
- B. Use customer-managed encryption keys (CMEK).
- C. Use customer-supplied encryption keys (CSEK).
- D. Use customer-supplied encryption keys (CSEK) for the sensitive data and customer-managed encryption keys (CMEK) for the less sensitive data.

Answer: A

Explanation:

A. Use Google-managed encryption keys (GMEK).

Google-managed encryption keys (GMEK) provide automatic encryption for your data without requiring you to generate, store, or manage encryption keys. Google handles the entire encryption and key management process for you.

This approach ensures robust security while minimizing the operational burden, which aligns with your need to avoid managing encryption keys.

Question: 20

CertyIQ

You work for a financial organization that stores transaction data in BigQuery. Your organization has a regulatory **requirement to retain data for a minimum of seven years for auditing purposes**. You need to ensure that the **data is**

retained for seven years using an efficient and cost-optimized approach. What should you do?

- A. Create a partition by transaction date, and set the partition expiration policy to seven years.
- B. Set the table-level retention policy in BigQuery to seven years.
- C. Set the dataset-level retention policy in BigQuery to seven years.
- D. Export the BigQuery tables to Cloud Storage daily, and enforce a lifecycle management policy that has a seven-year retention rule.

Answer: B

Explanation:

B. Set the table-level retention policy in BigQuery to seven years.

Setting a table-level retention policy in BigQuery ensures that the data is automatically retained for the required seven years without accidental deletion. It aligns perfectly with regulatory compliance requirements.

This approach is efficient because it is enforced directly within BigQuery, reducing operational overhead.

It is cost-optimized as you retain only what is necessary for compliance, avoiding the need for additional data exports and external storage.

Question: 21

CertyIQ

You need to create a weekly aggregated sales report based on a large volume of data. You want to use Python to design an efficient process for generating this report. What should you do?

- A. Create a Cloud Run function that uses NumPy. Use Cloud Scheduler to schedule the function to run once a week.
- B. Create a Colab Enterprise notebook and use the bigframes.pandas library. Schedule the notebook to execute once a week.
- C. Create a Cloud Data Fusion and Wrangler flow. Schedule the flow to run once a week.
- D. Create a Dataflow directed acyclic graph (DAG) coded in Python. Use Cloud Scheduler to schedule the code to run once a week.

Answer: B

Explanation:

Create a Colab Enterprise notebook and use the bigframes.pandas library. Schedule the notebook to execute once a week.

Colab Enterprise notebooks provide an interactive environment for Python-based data analysis, making them a good choice for rapid prototyping or development in environments where collaboration is key.

The bigframes.pandas library simplifies data manipulation and aggregation tasks, making it easier to handle structured datasets with Pythonic tools.

If the dataset is not excessively large and the operational requirements are not too stringent, this approach can be implemented quickly without needing advanced infrastructure setup.

Scheduling the notebook to run weekly ensures that the report generation is automated.

Question: 22

CertyIQ

Your organization has decided to move their on-premises Apache Spark-based workload to Google Cloud. You want to be able to **manage the code without needing to provision and manage your own cluster**. What should you do?

- A. Migrate the Spark jobs to Dataproc Serverless.
- B. Configure a Google Kubernetes Engine cluster with Spark operators, and deploy the Spark jobs.
- C. Migrate the Spark jobs to Dataproc on Google Kubernetes Engine.
- D. Migrate the Spark jobs to Dataproc on Compute Engine.

Answer: A

Explanation:

A. Migrate the Spark jobs to Dataproc Serverless.

No Cluster Management: Dataproc Serverless eliminates the need to provision and manage your own infrastructure, allowing you to focus solely on your code.

Scalability: It automatically scales resources based on the requirements of your workload, ensuring optimal performance without manual intervention.

Cost Efficiency: You pay only for the resources you use during job execution, which makes it a cost-effective solution.

Seamless Migration: Dataproc Serverless supports Spark workloads, ensuring a smooth transition from your on-premises setup.

Question: 23

CertyIQ

You are developing a data ingestion pipeline to load small CSV files into BigQuery from Cloud Storage. You want to **load these files upon arrival to minimize data latency**. You want to accomplish this with **minimal cost and maintenance**. What should you do?

- A. Use the bq command-line tool within a Cloud Shell instance to load the data into BigQuery.
- B. Create a Cloud Composer pipeline to load new files from Cloud Storage to BigQuery and schedule it to run every 10 minutes.
- C. Create a Cloud Run function to load the data into BigQuery that is triggered when data arrives in Cloud Storage.
- D. Create a Dataproc cluster to pull CSV files from Cloud Storage, process them using Spark, and write the results to BigQuery.

Answer: C

Explanation:

C. Create a Cloud Run function to load the data into BigQuery that is triggered when data arrives in Cloud Storage.

Event-driven automation: Cloud Run can be triggered automatically by Cloud Storage events (using Eventarc), ensuring that files are loaded into BigQuery as soon as they arrive, minimizing latency.

Serverless and cost-effective: Cloud Run is fully serverless, so you only pay when your function runs. There's no need to manage infrastructure, keeping costs and operational overhead minimal.

Scalability: Cloud Run can handle large numbers of events efficiently, ensuring that even if many files arrive simultaneously, they will be processed seamlessly.

Question: 24

CertyIQ

Your organization has a petabyte of application logs stored as **Parquet files** in Cloud Storage. You need to quickly perform a **one-time SQL-based analysis of the files and join them to data that already resides in BigQuery**. What should you do?

- A. Create a Dataproc cluster, and write a PySpark job to join the data from BigQuery to the files in Cloud Storage.
- B. Launch a Cloud Data Fusion environment, use plugins to connect to BigQuery and Cloud Storage, and use the SQL join operation to analyze the data.
- C. Create **external tables** over the files in Cloud Storage, and perform SQL joins to tables in BigQuery to analyze the data.
- D. Use the bq load command to load the Parquet files into BigQuery, and perform SQL joins to analyze the data.

Answer: C

Explanation:

C. Create external tables over the files in Cloud Storage, and perform SQL joins to tables in BigQuery to analyze the data.

External tables in BigQuery allow you to query data stored in Cloud Storage directly without having to load it into BigQuery, which makes this approach both time- and cost-efficient for a one-time analysis.

By creating an external table, you can use SQL to perform the required joins with existing BigQuery tables. This simplifies the process and avoids the additional overhead of transferring a petabyte of data into BigQuery.

This approach ensures minimal setup and leverages BigQuery's SQL capabilities efficiently.

Question: 25

CertyIQ

Your team is building several **data pipelines** that contain a **collection of complex tasks and dependencies that you want to execute on a schedule, in a specific order**. The tasks and dependencies consist of files in Cloud Storage, Apache Spark jobs, and data in BigQuery. You need to design a system that can **schedule and automate these data processing tasks using a fully managed approach**. What should you do?

- A. Use Cloud Scheduler to schedule the jobs to run.
- B. Use Cloud Tasks to schedule and run the jobs asynchronously.
- C. Create directed acyclic graphs (DAGs) in Cloud Composer. Use the appropriate operators to connect to Cloud Storage, Spark, and BigQuery.
- D. Create directed acyclic graphs (DAGs) in Apache Airflow deployed on Google Kubernetes Engine. Use the appropriate operators to connect to Cloud Storage, Spark, and BigQuery.

Answer: C

Explanation:

C. Create directed acyclic graphs (DAGs) in Cloud Composer. Use the appropriate operators to connect to Cloud Storage, Spark, and BigQuery.

Cloud Composer is a fully managed version of Apache Airflow provided by Google Cloud, specifically designed for orchestrating workflows involving complex dependencies and scheduling tasks in a specific order.

Directed Acyclic Graphs (DAGs) enable you to clearly define task dependencies and execution order, which is ideal for your scenario of managing multiple tasks and dependencies involving Cloud Storage, Spark, and BigQuery.

Cloud Composer integrates seamlessly with Google Cloud services like BigQuery and Cloud Storage, as well as Apache Spark jobs, using readily available operators.

It eliminates the need to manage the underlying infrastructure, reducing operational overhead while offering a scalable and reliable workflow orchestration system.

Question: 26

CertyIQ

You are responsible for managing Cloud Storage buckets for a research company. Your company has **well-defined data tiering and retention rules**. You need to **optimize storage costs** while **achieving your data retention needs**. What should you do?

- A. Configure the buckets to use the Archive storage class.
- B. Configure a lifecycle management policy on each bucket to downgrade the storage class and remove objects based on age.
- C. Configure the buckets to use the Standard storage class and enable Object Versioning.
- D. Configure the buckets to use the Autoclass feature.

Answer: B

Explanation:

B. Configure a lifecycle management policy on each bucket. Lifecycle Management is designed to automate data tiering (moving data to cheaper classes as it ages) and data retention (deleting objects based on age or other rules), directly addressing the requirements for cost optimization and rule-based data management. Option D (Autoclass) helps with tiering based on access but doesn't handle retention rules. Option A (Archive storage) is too simplistic and inflexible, not addressing tiering or retention rules properly. Option C (Standard + Versioning) increases costs and is counter to the goal of optimization. Therefore, Option B provides the most comprehensive and rule-driven approach for cost optimization and data lifecycle management in Cloud Storage.

Question: 27

CertyIQ

You are using your own data to demonstrate the capabilities of BigQuery to your organization's leadership team. You need to **perform a one-time load of the files stored on your local machine into BigQuery using as little effort as possible**. What should you do?

- A. Write and execute a Python script using the BigQuery Storage Write API library.
- B. Create a Dataproc cluster, copy the files to Cloud Storage, and write an Apache Spark job using the spark-bigquery-connector.
- C. Execute the bq load command on your local machine.
- D. Create a Dataflow job using the Apache Beam FileIO and BigQueryIO connectors with a local runner.

Answer: C

Explanation:

C. Execute the bq load command on your local machine.

The bq load command is the simplest and quickest way to load files stored on your local machine directly into BigQuery. It requires minimal setup and is perfect for a one-time load.

This method involves no additional infrastructure or programming, making it highly efficient for your demonstration purposes.

The command-line tool supports a variety of file formats (e.g., CSV, JSON, Avro) and allows you to specify schema and table details easily.

Question: 28**CertyIQ**

Your organization uses **Dataflow pipelines to process real-time financial transactions**. You discover that one of your Dataflow jobs has failed. You need to **troubleshoot the issue as quickly as possible**. What should you do?

- A. Set up a Cloud Monitoring dashboard to track key Dataflow metrics, such as data throughput, error rates, and resource utilization.
- B. Create a custom script to periodically poll the Dataflow API for job status updates, and send email alerts if any errors are identified.
- C. Navigate to the **Dataflow** Jobs page in the Google Cloud console. Use the job logs and worker logs to identify the error.
- D. Use the gcloud CLI tool to retrieve job metrics and logs, and analyze them for errors and performance bottlenecks.

Answer: C**Explanation:**

C. Navigate to the Dataflow Jobs page in the Google Cloud console. Use the job logs and worker logs to identify the error.

The Dataflow Jobs page in the Google Cloud console provides an intuitive interface to view job details, execution stages, and errors in real-time.

You can easily access job logs and worker logs, which are crucial for identifying the root cause of the failure.

This method is the fastest and most direct way to troubleshoot the issue, especially for real-time financial transaction processing where quick resolution is critical.

Question: 29**CertyIQ**

Your company uses Looker to generate and share reports with various stakeholders. You have a complex dashboard with several visualizations that needs to be delivered to **specific stakeholders on a recurring basis**, with **customized filters applied for each recipient**. You need an **efficient and scalable solution to automate the delivery of this customized dashboard**. You want to follow the **Google-recommended approach**. What should you do?

- A. Create a separate LookML model for each stakeholder with predefined filters, and schedule the dashboards using the Looker Scheduler.
- B. Create a script using the Looker Python SDK, and configure user attribute filter values. Generate a new scheduled plan for each stakeholder.
- C. Embed the Looker dashboard in a custom web application, and use the application's scheduling features to send the report with personalized filters.

D. Use the **Looker Scheduler** with a user attribute filter on the dashboard, and send the dashboard with personalized filters to each stakeholder based on their attributes.

Answer: D

Explanation:

D. Use Looker Scheduler with User Attribute filters. This is the most efficient and scalable approach as it directly utilizes Looker's built-in features to automate personalized dashboard delivery based on user attributes. Option A (Separate LookML models) is inefficient and unscalable due to model duplication. Option B (Python SDK scripting) is more complex than necessary and less efficient than using built-in features. Option C (Embedded Dashboard and Custom App) is overly complex and inefficient, re-implementing Looker's native functionalities. Therefore, Option D is the most straightforward, efficient, and aligned with Google's recommended best practices for Looker.

Question: 30

CertyIQ

You are predicting customer **churn for a subscription-based service**. You have a **50 PB** historical customer dataset in BigQuery that includes demographics, subscription information, and engagement metrics. You want to **build a churn prediction model with minimal overhead**. You want to follow the **Google-recommended approach**. What should you do?

- A. Export the data from BigQuery to a local machine. Use scikit-learn in a Jupyter notebook to build the churn prediction model.
- B. Use Dataproc to create a Spark cluster. Use the Spark MLlib within the cluster to build the churn prediction model.
- C. Create a Looker dashboard that is connected to BigQuery. Use LookML to predict churn.
- D. Use the BigQuery Python client library in a Jupyter notebook to query and preprocess the data in BigQuery. Use the **CREATE MODEL** statement in BigQueryML to train the churn prediction model.

Answer: D

Explanation:

D. Use BigQuery Python client and BigQueryML. BigQueryML enables in-database model training, eliminating data movement and minimizing overhead. This aligns with Google's best practices for BigQuery data. Option A (Local scikit-learn) is impractical due to the dataset size. Option B (Dataproc/Spark) introduces unnecessary data movement and cluster management overhead. Option C (Looker) is for BI, not ML model development. Therefore, Option D is the optimal choice for efficiency, scalability, and adherence to Google's recommendations for BigQuery-based machine learning.

Question: 31

CertyIQ

You are a data analyst at your organization. You have been given a BigQuery dataset that includes customer information. The dataset **contains inconsistencies and errors, such as missing values, duplicates, and formatting issues**. You need to **effectively and quickly clean the data**. What should you do?

- A. Develop a Dataflow pipeline to read the data from BigQuery, perform data quality rules and transformations, and write the cleaned data back to BigQuery.
- B. Use Cloud Data Fusion to create a data pipeline to read the data from BigQuery, perform data quality transformations, and write the clean data back to BigQuery.
- C. Export the data from BigQuery to CSV files. Resolve the errors using a spreadsheet editor, and re-import the cleaned data into BigQuery.

D. Use **BigQuery's built-in functions** to perform data quality transformations.

Answer: D

Explanation:

D. Use BigQuery's built-in functions to perform data quality transformations.

BigQuery's built-in functions are specifically designed to handle data quality tasks efficiently. Functions like IFNULL(), COALESCE() for missing values, SAFE_CAST() for correcting data types, DISTINCT or ROW_NUMBER() for deduplication, and string manipulation functions can address most common inconsistencies in your dataset.

This approach allows you to clean the data directly in BigQuery without requiring additional tools or workflows, minimizing effort and complexity.

It's the fastest and most straightforward solution for data cleaning tasks, as you can execute SQL transformations directly on the dataset.

Question: 32

CertyIQ

Your organization has several datasets in their data warehouse in BigQuery. Several analyst teams in different departments use the datasets to run queries. Your organization is concerned about the **variability of their monthly BigQuery costs**. You need to identify a **solution that creates a fixed budget for costs** associated with the queries run by each department. What should you do?

- A. Create a custom quota for each analyst in BigQuery.
- B. Create a single reservation by using BigQuery editions. Assign all analysts to the reservation.
- C. Assign each analyst to a separate project associated with their department. Create a single reservation by using BigQuery editions. Assign all projects to the reservation.
- D. Assign each analyst to a separate project associated with their department. Create a single reservation for each department by using BigQuery editions. **Create assignments for each project in the appropriate reservation.**

Answer: D

Explanation:

D. Assign each analyst to a separate project associated with their department. Create a single reservation for each department by using BigQuery editions. Create assignments for each project in the appropriate reservation.

Separate projects for departments: Assigning analysts to department-specific projects enables clear tracking and allocation of costs per department.

Separate reservations for departments: By creating a reservation for each department, you can allocate fixed capacity (e.g., slots) to match their expected usage. This ensures cost predictability for each department since you're controlling the resources they can use.

Assignments for projects to reservations: Associating each department's projects with their respective reservations ensures that the reserved capacity is utilized only by the department it is intended for.

Question: 33

CertyIQ

You manage a web application that stores data in a Cloud SQL database. You need to improve the read performance of the application by **offloading read traffic from the primary database instance**. You want to implement a solution that minimizes effort and cost. What should you do?

- A. Use Cloud CDN to cache frequently accessed data.
- B. Store frequently accessed data in a Memorystore instance.
- C. Migrate the database to a larger Cloud SQL instance.
- D. Enable automatic backups, and create a read replica of the Cloud SQL instance.

Answer: D

Explanation:

D. Enable automatic backups, and create a read replica of the Cloud SQL instance.

Read replicas are designed to offload read traffic from the primary database instance, significantly improving read performance without affecting write operations.

This solution minimizes effort, as it is straightforward to configure within the Cloud SQL settings.

It is cost-effective because you pay only for the additional resources used by the read replica, avoiding the higher costs associated with migrating to a larger instance.

Question: 34

CertyIQ

Your organization plans to move their on-premises environment to Google Cloud. Your organization's network **bandwidth is less than 1 Gbps**. You need to move over **500 TB** of data to Cloud Storage securely, and only have a few days to move the data. What should you do?

- A. Request multiple **Transfer Appliances**, copy the data to the appliances, and ship the appliances back to Google Cloud to upload the data to Cloud Storage.
- B. Connect to Google Cloud using VPN. Use Storage Transfer Service to move the data to Cloud Storage.
- C. Connect to Google Cloud using VPN. Use the `gcloud storage` command to move the data to Cloud Storage.
- D. Connect to Google Cloud using Dedicated Interconnect. Use the `gcloud storage` command to move the data to Cloud Storage.

Answer: A

Explanation:

A. Request multiple Transfer Appliances, copy the data to the appliances, and ship the appliances back to Google Cloud to upload the data to Cloud Storage.

Transfer Appliances are specifically designed for securely and efficiently transferring large volumes of data (like 500 TB) to Google Cloud when the available network bandwidth is insufficient.

With a bandwidth of less than 1 Gbps, transferring such a large amount of data over the network (options B, C, or D) would take far too long and is not practical, especially with a tight deadline.

The Transfer Appliance process is straightforward: you request the appliance, load your data onto it, and ship it back to Google Cloud. Once Google Cloud receives the appliance, your data is uploaded to Cloud Storage.

Question: 35

CertyIQ

Your organization uses a BigQuery table that is **partitioned by ingestion time**. You need to **remove data that is older than one year** to reduce your organization's storage costs. You want to use the most efficient approach while **minimizing cost**. What should you do?

- A. Create a scheduled query that periodically runs an update statement in SQL that sets the "deleted" column to "yes" for data that is more than one year old. Create a view that filters out rows that have been marked deleted.
- B. Create a view that filters out rows that are older than one year.
- C. Require users to specify a partition filter using the alter table statement in SQL.
- D. Set the table partition expiration period to one year using the **ALTER TABLE** statement in SQL.

Answer: D

Explanation:

D. Set the table partition expiration period to one year using the ALTER TABLE statement in SQL. Partition expiration is a built-in BigQuery feature specifically designed to automatically delete old partitions, directly reducing storage costs and requiring minimal effort. Options A and B (soft delete and views) do not reduce storage costs; they only filter or mark data without removing it. Option C (requiring partition filters) is about query optimization, not data removal or storage cost reduction. Therefore, Option D is the only option that directly and effectively addresses the requirement of removing old data to minimize storage costs in a BigQuery partitioned table.

Question: 36

CertyIQ

Your company is migrating their **batch transformation pipelines** to Google Cloud. You need to choose a solution that supports **programmatic transformations using only SQL**. You also want the technology to support **Git integration for version control of your pipelines**. What should you do?

- A. Use Cloud Data Fusion pipelines.
- B. Use **Dataform** workflows.
- C. Use Dataflow pipelines.
- D. Use Cloud Composer operators.

Answer: B

Explanation:

B. Use Dataform workflows.

Dataform workflows are purpose-built for SQL-based transformations and allow you to define data pipelines entirely using SQL, making them an excellent choice for programmatic transformations without needing additional programming expertise.

Dataform offers built-in Git integration, enabling you to version control your pipelines seamlessly. This is crucial for maintaining the reliability and traceability of your workflows, especially in collaborative environments.

Dataform integrates easily with BigQuery, making it efficient for batch transformation pipelines in Google Cloud.

Question: 37

CertyIQ

You manage a BigQuery table that is used for critical end-of-month reports. The table is updated weekly with new sales data. You want to prevent data loss and reporting issues if the table is accidentally deleted. What should you do?

- A. Configure the time travel duration on the table to be exactly seven days. On deletion, re-create the deleted table solely from the time travel data.
- B. Schedule the creation of a new snapshot of the table once a week. On deletion, re-create the deleted table using the snapshot and time travel data.
- C. Create a clone of the table. On deletion, re-create the deleted table by copying the content of the clone.
- D. Create a view of the table. On deletion, re-create the deleted table from the view and time travel data.

Answer: B

Explanation:

B. Schedule the creation of a new snapshot of the table once a week. On deletion, re-create the deleted table using the snapshot and time travel data.

Snapshots: BigQuery table snapshots create a point-in-time copy of your table, providing a highly reliable mechanism to restore the table in case of accidental deletion. Snapshots remain independent of the original table and retain the data for as long as required, offering a safeguard beyond the time travel period.

Time Travel: BigQuery's time travel feature allows you to access table data from up to the last seven days. Together with snapshots, this ensures you can restore data even beyond the seven-day window by combining both mechanisms.

Question: 38

CertyIQ

Your organization sends IoT event data to a Pub/Sub topic. Subscriber applications read and perform transformations on the messages before storing them in the data warehouse. During particularly busy times when more data is being written to the topic, you notice that the subscriber applications are not acknowledging messages within the deadline. You need to modify your pipeline to handle these activity spikes and continue to process the messages. What should you do?

- A. Retry messages until they are acknowledged.
- B. Implement flow control on the subscribers.
- C. Forward unacknowledged messages to a dead-letter topic.
- D. Seek back to the last acknowledged message.

Answer: B

Explanation:

B. Implement flow control on the subscribers.

Flow control allows subscriber applications to handle spikes in data volume by controlling the rate at which they receive messages from the Pub/Sub topic. This prevents the subscriber from becoming overwhelmed and ensures a steady processing rate.

With flow control, you can configure limits such as the number of outstanding messages or the total size of unprocessed messages. When these limits are reached, the subscriber will pause message delivery temporarily, allowing it to catch up.

Question: 39

CertyIQ

You have millions of customer feedback records stored in BigQuery. You want to **summarize the data by using the large language model (LLM) Gemini**. You need to plan and execute this analysis using the **most efficient approach**. What should you do?

- A. Query the BigQuery table from within a Python notebook, use the Gemini API to summarize the data within the notebook, and store the summaries in BigQuery.
- B. Use a BigQuery ML model to pre-process the text data, export the results to Cloud Storage, and use the Gemini API to summarize the pre-processed data.
- C. Create a BigQuery Cloud resource connection to a remote model in Vertex AI, and use Gemini to summarize the data.
- D. Export the raw BigQuery data to a CSV file, upload it to Cloud Storage, and use the Gemini API to summarize the data.

Answer: C

Explanation:

C. Create a BigQuery Cloud resource connection to a remote model in Vertex AI, and use Gemini to summarize the data.

Integration with Vertex AI: Using Vertex AI allows you to directly connect BigQuery data to Gemini, avoiding the need to export or manually process large datasets. This seamless integration reduces latency and complexity in the workflow.

Efficiency and Scalability: BigQuery and Vertex AI are designed to handle large-scale data processing and AI model execution, making them well-suited for summarizing millions of feedback records efficiently.

Streamlined Workflow: By setting up a Cloud resource connection, you eliminate the need for intermediary steps, such as exporting data or managing files manually.

Question: 40

CertyIQ

You are working on a data pipeline that will **validate and clean incoming data** before loading it into BigQuery for **real-time analysis**. You want to ensure that the data validation and cleaning is performed **efficiently and can handle high volumes of data**. What should you do?

- A. Write custom scripts in Python to validate and clean the data outside of Google Cloud. Load the cleaned data into BigQuery.
- B. Use Cloud Run functions to trigger data validation and cleaning routines when new data arrives in Cloud Storage.
- C. Use **Dataflow** to create a streaming pipeline that includes validation and transformation steps.
- D. Load the raw data into BigQuery using Cloud Storage as a staging area, and use SQL queries in BigQuery to validate and clean the data.

Answer: C

Explanation:

C. Use Dataflow to create a streaming pipeline that includes validation and transformation steps.

Dataflow is built for high-performance, distributed data processing, making it ideal for handling high volumes of incoming data in real time.

With Apache Beam, which powers Dataflow, you can programmatically implement data validation and

transformation steps as part of a robust streaming pipeline.

Dataflow ensures scalability and can process large-scale data seamlessly, maintaining efficiency during spikes in data volume.

It integrates well with BigQuery, enabling you to load cleaned and validated data efficiently for real-time analysis.

Question: 41

CertyIQ

Your organization needs to implement **near real-time analytics for thousands of events arriving each second** in Pub/Sub. The incoming messages require transformations. You need to **configure a pipeline** that **processes, transforms, and loads** the data into BigQuery while **minimizing development time**. What should you do?

- A. Use a Google-provided **Dataflow template** to process the Pub/Sub messages, perform transformations, and write the results to BigQuery.
- B. Create a Cloud Data Fusion instance and configure Pub/Sub as a source. Use Data Fusion to process the Pub/Sub messages, perform transformations, and write the results to BigQuery.
- C. Load the data from Pub/Sub into Cloud Storage using a Cloud Storage subscription. Create a Dataproc cluster, use PySpark to perform transformations in Cloud Storage, and write the results to BigQuery.
- D. Use Cloud Run functions to process the Pub/Sub messages, perform transformations, and write the results to BigQuery.

Answer: A

Explanation:

A. Use a Google-provided Dataflow template to process the Pub/Sub messages, perform transformations, and write the results to BigQuery.

Minimizes development time: Google-provided Dataflow templates offer pre-built pipelines for common tasks, such as processing Pub/Sub messages and loading data into BigQuery. These templates significantly reduce the time and effort needed to build and deploy a custom pipeline.

Real-time scalability: Dataflow is designed for real-time, high-throughput data processing. It can handle thousands of events per second efficiently.

Transformation support: You can extend or customize the pipeline to include necessary transformations using Apache Beam, which is the underlying framework of Dataflow.

Integration with BigQuery: Dataflow integrates seamlessly with Pub/Sub and BigQuery, allowing for efficient data ingestion and real-time analytics.

Question: 42

CertyIQ

Your organization needs to store historical customer order data. The data will only be accessed **once a month** for analysis and must be readily available **within a few seconds when it is accessed**. You need to choose a storage class that **minimizes storage costs** while ensuring that the data **can be retrieved quickly**. What should you do?

- A. Store the data in Cloud Storage using **Nearline** storage.
- B. Store the data in Cloud Storage using Coldline storage.
- C. Store the data in Cloud Storage using Standard storage.
- D. Store the data in Cloud Storage using Archive storage.

Answer: A

Explanation:

A. Store the data in Cloud Storage using Nearline storage. Nearline provides the optimal balance of minimizing storage costs while ensuring data is readily available within a few seconds when accessed monthly. Coldline (B) is cheaper storage but has higher latency and retrieval costs, potentially making it slightly slower and more expensive for monthly access. Standard (C) is too expensive for infrequently accessed data. Archive (D) is far too slow for "seconds" retrieval, with latency in hours. Therefore, Nearline is the most suitable choice for balancing cost and performance for this specific use case.

Question: 43

CertyIQ

You have a **Dataflow pipeline** that processes **website traffic logs stored** in Cloud Storage and **writes the processed data to BigQuery**. You noticed that the **pipeline is failing intermittently**. You need to **troubleshoot** the issue. What should you do?

- A. Use Cloud Logging to identify error groups in the pipeline's logs. Use Cloud Monitoring to create a dashboard that tracks the number of errors in each group.
- B. Use Cloud Logging to create a chart displaying the pipeline's error logs. Use Metrics Explorer to validate the findings from the chart.
- C. Use **Cloud Logging to view error messages** in the pipeline's logs. Use **Cloud Monitoring to analyze** the pipeline's metrics, such as CPU utilization and memory usage.
- D. Use the Dataflow job monitoring interface to check the pipeline's status every hour. Use Cloud Profiler to analyze the pipeline's metrics, such as CPU utilization and memory usage.

Answer: C

Explanation:

C. Use Cloud Logging to view error messages in the pipeline's logs. Use Cloud Monitoring to analyze the pipeline's metrics, such as CPU utilization and memory usage.

Cloud Logging: Viewing the error messages in the pipeline's logs allows you to pinpoint specific errors or exceptions causing the intermittent failures. These logs provide crucial insights into the root cause of the issues.

Cloud Monitoring: Analyzing metrics like CPU utilization, memory usage, and disk I/O can help identify resource bottlenecks or performance anomalies that might contribute to the failures.

Question: 44

CertyIQ

Your organization's business analysts require near **real-time access to streaming data**. However, they are reporting that their **dashboard queries are loading slowly**. After investigating BigQuery query performance, you discover the slow dashboard queries **perform several joins and aggregations**. You need to improve the dashboard loading time and ensure that the dashboard data is as up-to-date as possible. What should you do?

- A. Disable BigQuery query result caching.
- B. Modify the schema to use parameterized data types.
- C. Create a scheduled query to calculate and store intermediate results.
- D. Create **materialized** views.

Answer: D

Explanation:

D. Create materialized views.

Materialized views precompute and store the results of a query, such as joins and aggregations, which are the primary reasons for your dashboard's slow loading time.

These precomputed results allow your dashboard queries to retrieve data quickly, significantly improving performance.

Materialized views in BigQuery are automatically updated whenever the underlying base table changes, ensuring the data remains as up-to-date as possible for near real-time analytics.

Question: 45

CertyIQ

You need to create a **data pipeline that streams event information from applications in multiple Google Cloud regions** into BigQuery for near real-time analysis. The data requires **transformation** before loading. You want to create the **pipeline using a visual interface**. What should you do?

- A. Push event information to a **Pub/Sub** topic. Create a **Dataflow** job using the **Dataflow job builder**.
- B. Push event information to a Pub/Sub topic. Create a Cloud Run function to subscribe to the Pub/Sub topic, apply transformations, and insert the data into BigQuery.
- C. Push event information to a Pub/Sub topic. Create a BigQuery subscription in Pub/Sub.
- D. Push event information to Cloud Storage, and create an external table in BigQuery. Create a BigQuery scheduled job that executes once each day to apply transformations.

Answer: A

Explanation:

A. Push event information to a Pub/Sub topic. Create a Dataflow job using the Dataflow job builder. This is because it directly addresses all requirements: streaming data from multiple regions using Pub/Sub, transformations within Dataflow, near real-time analysis with Dataflow streaming pipelines, and a visual interface using the Dataflow Job Builder. Option B (Cloud Run) is less ideal for robust streaming pipelines and visual creation. Option C (BigQuery Subscription) lacks data transformation. Option D (Cloud Storage/Scheduled Job) is a batch, daily process, not near real-time streaming.

Question: 46

CertyIQ

You work for an online retail company. Your company collects customer purchase data in CSV files and pushes them to Cloud Storage **every 10 minutes**. The data needs to be **transformed and loaded into BigQuery for analysis**. The transformation involves **cleaning the data, removing duplicates, and enriching it with product information from a separate table** in BigQuery. You need to **implement a low-overhead solution that initiates data processing as soon as the files are loaded into Cloud Storage**. What should you do?

- A. Use Cloud Composer sensors to detect files loading in Cloud Storage. Create a Dataproc cluster, and use a Composer task to execute a job on the cluster to process and load the data into BigQuery.
- B. Schedule a direct acyclic graph (DAG) in Cloud Composer to run hourly to batch load the data from Cloud Storage to BigQuery, and process the data in BigQuery using SQL.
- C. Use **Dataflow** to implement a streaming pipeline using an **OBJECT_FINALIZE** notification from **Pub/Sub** to read the data from Cloud Storage, perform the transformations, and write the data to BigQuery.
- D. Create a Cloud Data Fusion job to process and load the data from Cloud Storage into BigQuery. Create an

OBJECT_FINALIZE notification in Pub/Sub, and trigger a Cloud Run function to start the Cloud Data Fusion job as soon as new files are loaded.

Answer: C

Explanation:

C. Use Dataflow to implement a streaming pipeline using an OBJECT_FINALIZE notification from Pub/Sub to read the data from Cloud Storage, perform the transformations, and write the data to BigQuery.

Event-driven pipeline: Using an OBJECT_FINALIZE notification ensures that the pipeline starts processing immediately whenever a new file is uploaded, minimizing latency.

Dataflow for transformations: Dataflow is designed for real-time or streaming data processing. It efficiently handles data cleaning, deduplication, and enrichment at scale.

Pub/Sub integration: Pub/Sub serves as a reliable event mechanism, notifying the Dataflow pipeline to start processing upon file arrival in Cloud Storage.

Direct integration with BigQuery: Dataflow has built-in connectors for BigQuery, enabling seamless loading of transformed data.

Question: 47

CertyIQ

You work for a home insurance company. You are frequently asked to create and save risk reports with charts for specific areas using a publicly available **storm event dataset**. You want to be able to quickly **create and re-run risk reports when new data becomes available**. What should you do?

A. Export the storm event dataset as a CSV file. Import the file to Google Sheets, and use cell data in the worksheets to create charts.

B. Copy the storm event dataset into your BigQuery project. Use BigQuery Studio to query and visualize the data in Looker Studio.

C. Reference and query the storm event dataset using SQL in BigQuery Studio. Export the results to Google Sheets, and use cell data in the worksheets to create charts.

D. Reference and query the storm event dataset using SQL in a Colab Enterprise notebook. Display the table results and document with Markdown, and use Matplotlib to create charts.

Answer: B

Explanation:

B. Copy the storm event dataset into your BigQuery project. Use BigQuery Studio to query and visualize the data in Looker Studio.

BigQuery + Looker Studio Integration: Once the storm event dataset is copied to your BigQuery project, you can leverage BigQuery Studio to perform SQL queries efficiently and process the data.

Visualization in Looker Studio: Looker Studio provides an intuitive interface for creating visual reports, including charts, which are dynamically linked to your BigQuery dataset. This allows you to easily refresh and update risk reports as new data becomes available.

Question: 48

CertyIQ

Your company currently uses an on-premises **network file system (NFS)** and is migrating data to Google Cloud. You

want to be able to **control how much bandwidth is used** by the data migration while capturing **detailed reporting on the migration status**. What should you do?

- A. Use a Transfer Appliance.
- B. Use Cloud Storage FUSE.
- C. Use **Storage Transfer Service**.
- D. Use gcloud storage commands.

Answer: C

Explanation:

Storage Transfer Service is the most appropriate solution because it directly addresses both requirements: bandwidth control and detailed reporting, specifically for data migration to Google Cloud Storage from sources like NFS.

Question: 49

CertyIQ

You are a Looker analyst. You need to **add a new field to your Looker report** that generates SQL that will run against your company's database. You do **not have the Develop permission**. What should you do?

- A. Create a new field in the LookML layer, refresh your report, and select your new field from the field picker.
- B. Create a calculated field using the Add a field option in Looker Studio, and add it to your report.
- C. Create a table calculation from the field picker in Looker, and add it to your report.
- D. Create a custom field from the field picker in Looker, and add it to your report.

Answer: D

Explanation:

D. Create a custom field from the field picker in Looker, and add it to your report.

Custom fields allow you to add new fields to your Looker report without requiring Develop permissions, which is ideal in your scenario.

By creating a custom field directly from the field picker, you can define the logic needed (e.g., simple calculations or SQL snippets) to generate the desired data and immediately include it in your report.

This process doesn't require modifications to the LookML layer, making it accessible for analysts with restricted permissions.

Question: 50

CertyIQ

Your organization's ecommerce website collects user activity logs using a Pub/Sub topic. Your organization's leadership team wants a **dashboard that contains aggregated user engagement metrics**. You need to create a solution that **transforms the user activity logs into aggregated metrics**, while ensuring that the **raw data can be easily queried**. What should you do?

- A. Create a Dataflow subscription to the Pub/Sub topic, and transform the activity logs. Load the transformed data into a BigQuery table for reporting.
- B. Create an event-driven Cloud Run function to trigger a data transformation pipeline to run. Load the transformed activity logs into a BigQuery table for reporting.
- C. Create a Cloud Storage subscription to the Pub/Sub topic. Load the activity logs into a bucket using the Avro

file format. Use Dataflow to transform the data, and load it into a BigQuery table for reporting.

D. Create a BigQuery subscription to the Pub/Sub topic, and load the activity logs into the table. Create a materialized view in BigQuery using SQL to transform the data for reporting

Answer: D

Explanation:

D. Create a BigQuery subscription to the Pub/Sub topic, and load the activity logs into the table. Create a materialized view in BigQuery using SQL to transform the data for reporting.

Google Cloud provides a Pub/Sub to BigQuery subscription, allowing you to stream data directly into BigQuery without needing Dataflow or Cloud Run.

This is the most efficient and serverless approach for real-time analytics.

Once the raw user activity logs are stored in BigQuery, you can:

Create a materialized view to aggregate user engagement metrics in real-time.

Query the raw logs easily without complex ETL pipelines.

Reduce operational overhead by using built-in BigQuery transformations.

Real-time Data Ingestion

Pub/Sub directly streams logs into BigQuery without needing an additional service.

Eliminates the need for Dataflow, Cloud Storage, or Cloud Run for data ingestion.

Efficient Data Transformation

BigQuery Materialized Views automatically update aggregated metrics.

No need for external ETL tools like Dataflow or Cloud Run.

Cost-Effective & Serverless

No extra compute costs (unlike Dataflow or Cloud Run).

Uses BigQuery's pay-as-you-go model, reducing infrastructure complexity.

Question: 51

CertyIQ

You are constructing a data pipeline to process sensitive customer data stored in a Cloud Storage bucket. You need to ensure that this data remains accessible, even in the event of a single-zone outage. What should you do?

- A. Set up a Cloud CDN in front of the bucket.
- B. Enable Object Versioning on the bucket.
- C. Store the data in a multi-region bucket.
- D. Store the data in Nearline storage.

Answer: C

Explanation:

C. Store the data in a multi-region bucket.

Multi-region buckets are designed for high availability and resilience. They store data redundantly across multiple geographic locations, ensuring that your data remains accessible even if a single zone experiences an outage.

This approach is particularly suitable for sensitive customer data that needs to remain available at all times, aligning with your pipeline's requirements.

Question: 52

CertyIQ

Your retail company collects customer data from various sources:

Online transactions: Stored in a MySQL database

Customer feedback: Stored as text files on a company server

Social media activity: Streamed in real-time from social media platforms

You are designing a data pipeline to extract this data. Which Google Cloud storage system(s) should you select for further analysis and ML model training?

- A.1. Online transactions: Cloud Storage
- 2. Customer feedback: Cloud Storage
- 3. Social media activity: Cloud Storage
- B.1. Online transactions: BigQuery
- 2. Customer feedback: Cloud Storage
- 3. Social media activity: BigQuery
- C.1. Online transactions: Bigtable
- 2. Customer feedback: Cloud Storage
- 3. Social media activity: CloudSQL for MySQL
- D.1. Online transactions: Cloud SQL for MySQL
- 2. Customer feedback: BigQuery
- 3. Social media activity: Cloud Storage

Answer: B

Explanation:

B.

1. Online transactions → BigQuery

2. Customer feedback → Cloud Storage

3. Social media activity → BigQuery.

Online Transactions → BigQuery

Online transactions are structured, relational data (previously stored in MySQL).

BigQuery is optimized for analytics on structured data.

Supports SQL-based queries for easy analysis.

Highly scalable and can handle large datasets efficiently.

Customer Feedback → Cloud Storage.

Feedback is stored as text files, which is unstructured data.

Cloud Storage is cost-effective and can handle large, unstructured datasets.

Easily integrates with Dataflow, BigQuery, and AI tools for further processing.

Social Media Activity → BigQuery.

Social media activity is streamed in real-time → BigQuery supports real-time ingestion using Pub/Sub & BigQuery streaming API.

Ideal for aggregations, trend analysis, and ML training.

Question: 53

CertyIQ

Your company uses Looker as its primary business intelligence platform. You want to use LookML to visualize the profit margin for each of your company's products in your Looker Explores and dashboards. You need to implement a solution quickly and efficiently. What should you do?

- A. Create a derived table that pre-calculates the profit margin for each product, and include it in the Looker model.
- B. Define a new measure that calculates the profit margin by using the existing revenue and cost fields.
- C. Create a new dimension that categorizes products based on their profit margin ranges (e.g., high, medium, low).
- D. Apply a filter to only show products with a positive profit margin.

Answer: B

Explanation:

B. Define a new measure. Option B is best: Measures are for on-the-fly calculations in Looker, quick & efficient. Option A (Derived table) is incorrect: Slower, pre-calculation not needed. Option C (New dimension) is incorrect: Dimensions are for categories, not metrics. Option D (Apply filter) is incorrect: Filter restricts data, doesn't calculate margin. Therefore, Option B, new measure, is the fastest, most efficient LookML solution.

Question: 54

CertyIQ

You are a data analyst working with sensitive customer data in BigQuery. You need to ensure that only authorized personnel within your organization can query this data, while following the principle of least privilege. What should you do?

- A. Enable access control by using IAM roles.
- B. Encrypt the data by using customer-managed encryption keys (CMEK).
- C. Update dataset privileges by using the SQL GRANT statement.
- D. Export the data to Cloud Storage, and use signed URLs to authorize access.

Answer: A

Explanation:

A. Enable access control by using IAM roles. Option A is optimal because IAM roles are the standard Google Cloud method for managing access, directly enabling least privilege for BigQuery. Option B (CMEK) is incorrect because encryption secures data but doesn't control access. Option C (SQL GRANT) is less preferred than IAM for broader GCP access management. Option D (Export and signed URLs) is incorrect as it's complex, less secure, and not for controlling BigQuery query access. Therefore, Option A is the most direct and secure way to manage access to sensitive BigQuery data.

Question: 55

CertyIQ

Your organization stores **highly personal data** in BigQuery and needs to comply with **strict data privacy regulations**. You need to ensure that **sensitive data values are rendered unreadable whenever an employee leaves the organization**. What should you do?

- A. Use **AEAD** functions and delete keys when employees leave the organization.
- B. Use dynamic data masking and revoke viewer permissions when employees leave the organization.
- C. Use customer-managed encryption keys (CMEK) and delete keys when employees leave the organization.
- D. Use column-level access controls with policy tags and revoke viewer permissions when employees leave the organization.

Answer: A

Explanation:

A. Use AEAD functions and delete keys when employees leave the organization. Option A is best because AEAD encryption makes data values unreadable by design when keys are deleted, directly meeting the requirement. Option B (Dynamic masking) is incorrect because masking only hides data, not making it unreadable at the storage level. Option C (CMEK) is less ideal than AEAD for this specific need, as it's broader encryption and potentially more disruptive. Option D (Column-level access control) is incorrect because it restricts access, but data remains readable for authorized users, not inherently unreadable. Therefore, Option A is the most precise solution for rendering data values unreadable upon employee departure.

Question: 56

CertyIQ

You used BigQuery ML to build a customer purchase propensity model **six months ago**. You want to **compare** the **current serving data** with the **historical serving data** to determine **whether you need to retrain the model**. What should you do?

- A. Compare the two different models.
- B. Evaluate the data skewness.
- C. Evaluate **data drift**.
- D. Compare the confusion matrix.

Answer: C

Explanation:

C. Evaluate data drift.

Data drift refers to changes in the data distribution of the serving data (current input) compared to the historical data (used during model training). Significant data drift can indicate that the model is no longer performing optimally and may need retraining.

By evaluating data drift, you can detect if the characteristics of the input data have shifted over time, which may affect the model's predictions.

Question: 57

CertyIQ

Your company uses Looker to visualize and analyze sales data. You need to create a dashboard that displays sales

metrics, such as sales by region, product category, and time period. Each metric relies on its own set of attributes distributed across several tables. You need to provide users the ability to filter the data by specific sales representatives and view individual transactions. You want to follow the Google-recommended approach. What should you do?

- A. Create multiple Explores, each focusing on each sales metric. Link the Explores together in a dashboard using drill-down functionality.
- B. Use BigQuery to create multiple materialized views, each focusing on a specific sales metric. Build the dashboard using these views.
- C. Create a single Explore with all sales metrics. Build the dashboard using this Explore.
- D. Use Looker's custom visualization capabilities to create a single visualization that displays all the sales metrics with filtering and drill-down functionality.

Answer: C

Explanation:

C. Create a single Explore with all sales metrics. Build the dashboard using this Explore.

Unified Explore: By creating a single Explore that includes all relevant sales metrics and their attributes, users can access a centralized source for building dashboards, enabling consistent filtering and drill-down functionality across metrics.

Efficient Data Model: Consolidating the attributes into a single Explore simplifies the data model, reducing the complexity for dashboard creation and user interactions.

Filtering and Drill-Down: Looker Explores inherently support filtering and drill-downs across related dimensions, making it straightforward for users to filter data by sales representatives and view individual transactions.

Question: 58

CertyIQ

Your company's ecommerce website collects product reviews from customers. The reviews are loaded as CSV files daily to a Cloud Storage bucket. The reviews are in multiple languages and need to be translated to Spanish. You need to configure a pipeline that is serverless, efficient, and requires minimal maintenance. What should you do?

- A. Load the data into BigQuery using Dataproc. Use Apache Spark to translate the reviews by invoking the Cloud Translation API. Set BigQuery as the sink.
- B. Use a Dataflow templates pipeline to translate the reviews using the Cloud Translation API. Set BigQuery as the sink.
- C. Load the data into BigQuery using a Cloud Run function. Use the BigQuery ML create model statement to train a translation model. Use the model to translate the product reviews within BigQuery.
- D. Load the data into BigQuery using a Cloud Run function. Create a BigQuery remote function that invokes the Cloud Translation API. Use a scheduled query to translate new reviews.

Answer: B

Explanation:

B. Dataflow template with Cloud Translation API. Option B is best because Dataflow templates are serverless, managed, and efficient for data pipelines like translation. Option A (Dataproc/Spark) is incorrect because Dataproc is not serverless and adds maintenance. Option C (Cloud Run/BigQuery ML) is incorrect because training a BigQuery ML model for translation is overly complex for this. Option D (Cloud Run/Remote Function) is incorrect because it adds unnecessary complexity with remote functions and scheduled queries. Therefore, Option B, Dataflow template, is the most streamlined and best-fit serverless solution.

Question: 59

CertyIQ

You have a **Dataproc cluster** that performs batch processing on data stored in Cloud Storage. You need to schedule a daily Spark job to generate a report that will be emailed to stakeholders. You need a **fully-managed solution** that is easy to implement and **minimizes complexity**. What should you do?

- A. Use Cloud Composer to orchestrate the Spark job and email the report.
- B. Use **Dataproc** workflow templates to define and schedule the Spark job, and to email the report.
- C. Use Cloud Run functions to trigger the Spark job and email the report.
- D. Use Cloud Scheduler to trigger the Spark job, and use Cloud Run functions to email the report.

Answer: B

Explanation:

B. Use Dataproc workflow templates to define and schedule the Spark job, and to email the report.

Dataproc workflow templates are designed for managing and automating workflows for Dataproc clusters. They allow you to define the sequence of jobs (e.g., your Spark job) and schedule them efficiently.

Built-in simplicity: The workflow template can be scheduled using Cloud Scheduler, ensuring that the job runs daily without requiring additional orchestration tools.

Emailing the report: You can configure a final step in the workflow to trigger a script or process that generates and emails the report to stakeholders. This keeps the entire solution within the Dataproc ecosystem, minimizing complexity.

Question: 60

CertyIQ

Your organization has highly sensitive data that gets **updated once a day** and is **stored across multiple datasets** in BigQuery. You need to provide a new data analyst **access to query specific data** in BigQuery while **preventing access to sensitive data**. What should you do?

- A. Grant the data analyst the BigQuery Job User IAM role in the Google Cloud project.
- B. Create a **materialized view** with the limited data in a new dataset. Grant the data analyst BigQuery Data Viewer IAM role in the dataset and the BigQuery Job User IAM role in the Google Cloud project.
- C. Create a new Google Cloud project, and copy the limited data into a BigQuery table. Grant the data analyst the BigQuery Data Owner IAM role in the new Google Cloud project.
- D. Grant the data analyst the BigQuery Data Viewer IAM role in the Google Cloud project.

Answer: B

Explanation:

B. Create a materialized view with the limited data in a new dataset. Grant the data analyst BigQuery Data Viewer IAM role in the dataset and the BigQuery Job User IAM role in the Google Cloud project.

Materialized View: By creating a materialized view, you can define specific data that the new analyst is allowed to query, ensuring sensitive data remains inaccessible. The materialized view focuses on the limited data you want to share while leaving sensitive datasets secure.

Access Control: Granting the BigQuery Data Viewer IAM role on the dataset containing the materialized view ensures the analyst can only view and query the specific data in that dataset.

BigQuery Job User Role: The BigQuery Job User IAM role allows the analyst to execute queries but does not grant direct access to any dataset. This ensures tighter control over data access.

Question: 61

CertyIQ

You are a database administrator managing sales transaction data by region stored in a BigQuery table. You need to ensure that each sales representative **can only see the transactions in their region**. What should you do?

- A. Add a policy tag in BigQuery.
- B. Create a **row-level access** policy.
- C. Create a data masking rule.
- D. Grant the appropriate IAM permissions on the dataset.

Answer: B

Explanation:

B. Create a row-level access policy.

Row-level access policies in BigQuery allow you to control access to specific rows within a table based on conditions, such as a user's region. This ensures that sales representatives only see the rows corresponding to their assigned region.

You can use row-level access policies to enforce fine-grained security without having to create multiple tables or datasets for each region.

Question: 62

CertyIQ

Your company's customer support audio files are stored in a **Cloud Storage bucket**. You plan to analyze the audio files' metadata and file content within BigQuery to create inference by using BigQuery ML. You need to create a corresponding table in BigQuery that represents the bucket containing the **audio files**. What should you do?

- A. Create an external table.
- B. Create a temporary table.
- C. Create a native table.
- D. **Create an object table.**

Answer: D

Explanation:

D. Create an object table.

Object tables in BigQuery are specifically designed to represent the contents of a Cloud Storage bucket. They automatically map each file in the bucket as a row in the table, enabling you to analyze metadata (e.g., file name, size, type) and integrate it with other BigQuery features.

Question: 63

CertyIQ

You work for a financial services company that handles highly sensitive data. Due to regulatory requirements, your company is required to have **complete and manual control of data encryption**. Which type of keys should you

recommend to use for data storage?

- A. Use customer-supplied encryption keys (CSEK).
- B. Use a dedicated third-party key management system (KMS) chosen by the company.
- C. Use Google-managed encryption keys (GMEK).
- D. Use customer-managed encryption keys (CMEK).

Answer: A

Explanation:

A. Use customer-supplied encryption keys (CSEK). Option A is best because CSEK provides complete manual control over keys, meeting strict regulatory needs. Option B (Third-party KMS) is incorrect because CSEK already provides control within Google Cloud. Option C (GMEK) is incorrect because Google manages keys, not the company. Option D (CMEK) is incorrect because keys are managed within Google KMS, not fully manual control. Therefore, Option A, CSEK, offers the most direct and complete manual key control.

Question: 64

CertyIQ

Your team needs to analyze large datasets stored in BigQuery to identify trends in user behavior. The analysis will involve **complex statistical calculations, Python packages, and visualizations**. You need to recommend **a managed collaborative environment to develop and share the analysis**. What should you recommend?

- A. Create a **Colab Enterprise notebook** and connect the notebook to BigQuery. Share the notebook with your team. Analyze the data and generate visualizations in Colab Enterprise.
- B. Create a statistical model by using BigQuery ML. Share the query with your team. Analyze the data and generate visualizations in Looker Studio.
- C. Create a Looker Studio dashboard and connect the dashboard to BigQuery. Share the dashboard with your team. Analyze the data and generate visualizations in Looker Studio.
- D. Connect Google Sheets to BigQuery by using Connected Sheets. Share the Google Sheet with your team. Analyze the data and generate visualizations in Google Sheets.

Answer: A

Explanation:

A. Create a Colab Enterprise notebook and connect the notebook to BigQuery. Share the notebook with your team. Analyze the data and generate visualizations in Colab Enterprise. Option A is best because Colab Enterprise provides a managed, collaborative Python environment directly connected to BigQuery, ideal for complex stats, Python packages, and visualizations. Option B (BigQuery ML & Looker Studio) is incorrect because BigQuery ML has limited statistical scope compared to Python, and Looker Studio is less suited for code-based analysis. Option C (Looker Studio dashboard) is incorrect because Looker Studio excels at dashboards, not complex Python analysis or statistical coding. Option D (Google Sheets) is incorrect because Sheets is not designed for large datasets or complex analysis requiring Python packages. Therefore, Option A, Colab Enterprise, best suits the need for a managed Python-based collaborative analysis environment for BigQuery data.

Question: 65

CertyIQ

Your organization has several datasets in BigQuery. The datasets need to be **shared with your external partners** so that they can run SQL queries without needing to copy the data to their own projects. You have organized each partner's data in its own BigQuery dataset. Each partner should be able to access only their data. You want to

share the data while following **Google-recommended practices**. What should you do?

- A. Use **Analytics Hub** to create a listing on a private data exchange for each partner dataset. Allow each partner to subscribe to their respective listings.
- B. Create a Dataflow job that reads from each BigQuery dataset and pushes the data into a dedicated Pub/Sub topic for each partner. Grant each partner the pubsub.subscriber IAM role.
- C. Export the BigQuery data to a Cloud Storage bucket. Grant the partners the storage.objectUser IAM role on the bucket.
- D. Grant the partners the bigquery.user IAM role on the BigQuery project.

Answer: A

Explanation:

A. Use Analytics Hub to create a listing on a private data exchange for each partner dataset. Allow each partner to subscribe to their respective listings.

Analytics Hub: Analytics Hub allows you to securely share BigQuery datasets through a private data exchange. By creating a listing for each partner dataset, you ensure that partners can access only their respective data.

Ease of Access: Partners can subscribe to their respective listings, enabling them to run SQL queries directly on the shared datasets without the need to copy data to their own projects.

Security: This method aligns with Google-recommended practices by leveraging Analytics Hub to enforce secure and efficient data sharing while maintaining control over dataset access.

Question: 66

CertyIQ

Your organization has decided to migrate their existing enterprise data warehouse to BigQuery. The existing data pipeline tools **already support connectors to BigQuery**. You need to identify a data migration approach that **optimizes migration speed**. What should you do?

- A. Create a temporary file system to facilitate data transfer from the existing environment to Cloud Storage. Use Storage Transfer Service to migrate the data into BigQuery.
- B. Use the Cloud Data Fusion web interface to build data pipelines. Create a directed acyclic graph (DAG) that facilitates pipeline orchestration.
- C. Use the **existing data pipeline** tool's BigQuery connector to reconfigure the data mapping.
- D. Use the BigQuery Data Transfer Service to recreate the data pipeline and migrate the data into BigQuery.

Answer: C

Explanation:

C. Use the existing data pipeline tool's BigQuery connector. Option C is best because leveraging existing tools and their BigQuery connectors is generally the fastest migration path. Reconfiguring existing mappings is more direct than introducing new services or intermediate steps, optimizing migration speed. Option A (Temporary file system and Storage Transfer Service) is incorrect because adding Cloud Storage as an intermediary step slows down migration compared to direct transfer. Option B (Cloud Data Fusion) is incorrect because building new pipelines in Data Fusion adds complexity and time, less optimal for speed if connectors already exist. Option D (BigQuery Data Transfer Service) is incorrect because Data Transfer Service is for scheduled transfers, not necessarily the fastest for a one-time bulk migration compared to direct connector usage. Therefore, Option C, using existing connectors, is the most direct and speed-optimized approach.

Question: 67

CertyIQ

Your organization uses scheduled queries to perform transformations on data stored in BigQuery. You discover that **one of your scheduled queries has failed**. You need to **troubleshoot the issue as quickly as possible**. What should you do?

- A. Navigate to the Logs Explorer page in Cloud Logging. Use filters to find the failed job, and analyze the error details.
- B. Set up a log sink using the gcloud CLI to export BigQuery audit logs to BigQuery. Query those logs to identify the error associated with the failed job ID.
- C. Request access from your admin to the BigQuery information_schema. Query the jobs view with the failed job ID, and analyze error details.
- D. Navigate to the **Scheduled queries page** in the Google Cloud console. Select the failed job, and analyze the error details.

Answer: D

Explanation:

D. Navigate to the Scheduled queries page in the Google Cloud console. Select the failed job, and analyze the error details.

When troubleshooting a failed scheduled query in BigQuery, the fastest and most direct method is to check the Scheduled Queries page in the Google Cloud console. This page provides:

Error messages & details related to the failed query.

Job execution history to identify patterns in failures.

Query logs to debug issues immediately.

Question: 68

CertyIQ

Your team uses the Google Ads platform to visualize metrics. You want to export the data to BigQuery to get more granular insights. You need to **execute a one-time transfer of historical data and automatically update data daily**. You want a solution that is **low-code, serverless, and requires minimal maintenance**. What should you do?

- A. Export the historical data to BigQuery by using BigQuery Data Transfer Service. Use Cloud Composer for daily automation.
- B. Export the historical data to Cloud Storage by using Storage Transfer Service. Use Pub/Sub to trigger a Dataflow template that loads data for daily automation.
- C. Export the historical data as a CSV file. Import the file into BigQuery for analysis. Use Cloud Composer for daily automation.
- D. Export the historical data **to BigQuery** by using BigQuery Data Transfer Service. Use BigQuery Data Transfer Service for daily automation.

Answer: D

Explanation:

D. Export the historical data to BigQuery by using BigQuery Data Transfer Service. Use BigQuery Data Transfer Service for daily automation.

The BigQuery Data Transfer Service (BQ DTS) is a serverless, low-code solution that allows you to seamlessly import data from Google Ads into BigQuery. It supports:

One-time transfer of historical data.

Automatic daily updates without manual intervention.

Fully managed service, minimizing maintenance and operational overhead.

Question: 69

CertyIQ

Your organization has a BigQuery dataset that contains **sensitive employee information** such as salaries and performance reviews. The payroll specialist in the HR department **needs to have continuous access** to aggregated performance data, but they **do not need continuous access to other sensitive data**. You need to grant the payroll specialist access to the performance data without granting them access to the entire dataset using the **simplest and most secure approach**. What should you do?

- A. Use **authorized views** to share query results with the payroll specialist.
- B. Create row-level and column-level permissions and policies on the table that contains performance data in the dataset. Provide the payroll specialist with the appropriate permission set.
- C. Create a table with the aggregated performance data. Use table-level permissions to grant access to the payroll specialist.
- D. Create a SQL query with the aggregated performance data. Export the results to an Avro file in a Cloud Storage bucket. Share the bucket with the payroll specialist.

Answer: A

Explanation:

A. Use authorized views to share query results with the payroll specialist.

The simplest and most secure way to grant access to only the aggregated performance data without exposing sensitive employee information (like salaries) is by using Authorized Views in BigQuery.

Authorized Views allow you to:

Control access at the query level rather than granting access to the entire dataset.

Expose only the necessary fields and aggregated data without revealing raw sensitive information.

Ensure data security and compliance by preventing unauthorized access to raw tables.

Question: 70

CertyIQ

You work for a global financial services company that trades stocks **24/7**. You have a Cloud SQL for PostgreSQL user database. You need to identify a solution that ensures that the database is **continuously operational, minimizes downtime, and will not lose any data in the event of a zonal outage**. What should you do?

- A. Continuously back up the Cloud SQL instance to Cloud Storage. Create a Compute Engine instance with PostgreSQL in a different region. Restore the backup in the Compute Engine instance if a failure occurs.
- B. Create a read replica in another region. Promote the replica to primary if a failure occurs.
- C. Configure and create a high-availability Cloud SQL instance with the primary instance in zone A and a secondary instance in any zone other than zone A.
- D. Create a read replica in the same region but in a different zone.

Answer: C

Explanation:

C. Configure and create a high-availability Cloud SQL instance with the primary instance in zone A and a

secondary instance in any zone other than zone A.

High availability (HA) configuration: Cloud SQL's HA setup ensures continuous operation by maintaining a primary instance and a synchronous standby instance in different zones. If a zonal outage occurs, the standby is automatically promoted to primary, minimizing downtime and preventing data loss.

Automatic Failover: The HA configuration handles failover automatically, providing seamless recovery without manual intervention, which is critical for 24/7 operations.

Data Consistency: With synchronous replication between the primary and standby, the HA setup ensures zero data loss in the event of an outage.

Question: 71

CertyIQ

You are migrating data from a legacy on-premises **MySQL database to Google Cloud**. The database contains various tables with different data types and sizes, including large tables with millions of rows and transactional data. You need to **migrate this data while maintaining data integrity, and minimizing downtime and cost**. What should you do?

- A. Set up a Cloud Composer environment to orchestrate a custom data pipeline. Use a Python script to extract data from the MySQL database and load it to MySQL on Compute Engine.
- B. Export the MySQL database to CSV files, transfer the files to Cloud Storage by using Storage Transfer Service, and load the files into a Cloud SQL for MySQL instance.
- C. Use Database Migration Service to replicate the MySQL database to a Cloud SQL for MySQL instance.
- D. Use Cloud Data Fusion to migrate the MySQL database to MySQL on Compute Engine.

Answer: C

Explanation:

C. Use Database Migration Service to replicate the MySQL database to a Cloud SQL for MySQL instance.

Database Migration Service (DMS): DMS is specifically designed for migrating MySQL, PostgreSQL, and SQL Server databases to Cloud SQL. It ensures data integrity by supporting replication of both data and schema while minimizing downtime with near-real-time replication.

Cost-Effectiveness: DMS is serverless and fully managed, eliminating the need for maintaining additional infrastructure, reducing migration costs.

Supports Large Data Volumes: DMS can handle millions of rows and transactional data efficiently, making it suitable for your use case.

Minimizing Downtime: By using continuous replication, DMS allows for a smooth cutover with minimal disruption.

Question: 72

CertyIQ

You created a customer support application that sends several forms of data to Google Cloud. Your application is sending:

1. Audio files from phone interactions with support agents that will be **accessed during trainings**.
2. CSV files of users' personally identifiable information (PII) that will be **analyzed with SQL**.
3. A large volume of **small document files that will power other applications**.

You need to select the appropriate tool for each data type given the required use case, while following Google-recommended practices. Which should you choose?

- A.1. Cloud Storage
- 2. CloudSQL for PostgreSQL
- 3. Bigtable
- B.1. Filestore
- 2. Cloud SQL for PostgreSQL
- 3. Datastore
- C.1. Cloud Storage
- 2. BigQuery
- 3. Firestore
- D.1. Filestore
- 2. Bigtable
- 3. BigQuery

Answer: C

Explanation:

C.

Cloud Storage

BigQuery

Firestore

1. Audio files from phone interactions → Cloud Storage

Best for storing unstructured data like audio, video, and images.

Provides low-cost, durable storage with lifecycle management and security.

Can be easily accessed by training applications when needed.

2. CSV files with PII for SQL analysis → BigQuery

Best for structured data that requires fast SQL-based analysis.

BigQuery has built-in security features to handle PII data securely.

Supports batch loading of CSV files for large-scale analytics.

3. A large volume of small document files → Firestore

Firestore is a document-based NoSQL database, ideal for storing and retrieving structured documents at scale.

Designed for high availability and low-latency access, making it a great choice for powering applications.

Fully managed, serverless solution with automatic scaling.

Thank you

Thank you for being so interested in the premium exam material.
I'm glad to hear that you found it informative and helpful.

If you have any feedback or thoughts on the bumps, I would love to hear them.
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Best of Luck

You have worked hard to get to this point, and you are well-prepared for the exam
Keep your head up, stay positive, and go show that exam what you're made of!

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